Sanaul Haque

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Full Legal Name: Karan Aggarwal
Institution: Independent Researcher
Category: Junior
Class/Grade/Year: Class 12 Graduate / Gap Year
Country: India
Major (if senior): Not Applicable
Submission Date: October 4, 2026
Registered Email Address: karanaggarwal2024@gmail.com
Research Topic: Strategic Under-Pricing in Online Auctions: Seller Reputation and Bidder Competition

Title: Strategic Under-Pricing in Online Auctions: Testing the Moderating Roles of Seller Reputation and Bidder Competition

Research Problem

In online auctions, sellers choose an opening price but the final transaction price emerges from bidder competition. This creates a practical pricing problem: can a deliberately low opening price attract enough participation to produce a higher final price, or does it simply expose the seller to greater downside risk? Standard auction theory does not predict an automatic revenue advantage from a low starting price. Under benchmark conditions, second-price auction mechanisms and revenue-equivalence results imply that expected outcomes depend primarily on bidder valuations and participation rather than on the nominal opening price alone [1], [2]. Online markets, however, depart from these conditions through endogenous entry, last-minute bidding, heterogeneous products, and incomplete information about sellers [3]-[5]. A low opening price may therefore work only under particular market conditions. Two plausible mechanisms are especially important: seller reputation, which can affect buyer trust, and bidder competition, which can determine whether an initially low price develops into an intensive bidding contest. Research is needed to test these mechanisms together rather than treating a simple correlation between opening and final prices as evidence that under-pricing itself is effective.

Existing Literature

Vickrey [1] established the theoretical properties of second-price auctions, and Myerson [2] developed the framework of optimal auction design. These benchmarks clarify how bidding mechanisms shape revenue, but their assumptions do not fully describe online marketplaces, where participation is endogenous and sellers differ in credibility. Bajari and Hortaçsu [3] examine reserve prices and bidder entry in eBay auctions; Lucking-Reiley [4] uses field experiments to show that auction format can affect outcomes. Roth and Ockenfels [5] document strategic last-minute bidding, while Cabral and Hortaçsu [6] establish that feedback conveys information about seller quality. Together, these studies suggest that entry and trust can matter, but they do not by themselves establish whether the relationship between unusually low opening prices and final prices varies jointly with seller reputation and the number of competing bidders. This proposal brings those two potential moderators into one empirical framework. It does not assume that an observed association identifies the causal effect of lowering an opening price.

Research Question

To what extent is strategic under-pricing associated with final auction prices in online auctions, and how is this relationship moderated by seller reputation and bidder competition? The study will examine three linked questions: (1) whether low opening prices are associated with final auction prices after accounting for product and auction characteristics; (2) whether seller reputation changes this relationship; and (3) whether greater bidder competition changes this relationship.

Methodology

The proposed empirical extension will assemble an auction-level sample of at least 600 recently completed eBay listings across product categories with sufficient price variation. This target is intended to support within-category comparisons and estimation of interaction terms; it is a collection target, not a claim that statistical power has already been established. The sampling window, categories, eligibility criteria, and exclusions will be specified before analysis, rather than selecting auctions on the basis of their final prices. Each completed auction will be one observation. Publicly observable fields will include opening and final prices, category, auction duration, bid activity, seller positive-feedback percentage, and seller feedback count. Distinct-bidder information will be collected only where it can be reliably verified. No private buyer or seller information will be collected.

Strategic under-pricing will be defined relative to other auctions in the same product category, not by a universal monetary cutoff. The primary indicator will identify opening prices in the lowest category-specific quartile, with a bottom-10% indicator used as a sensitivity check. Because even a category can contain products of different values and conditions, the models will also consider observable product characteristics where available and assess continuous opening-price specifications. Reputation will be measured using positive-feedback percentage and the logarithm of feedback count. Competition will be represented by the number of distinct bidders when visible. The availability of that measure will be reported by category and auction characteristics; auctions without verifiable distinct-bidder counts will not be silently treated as having zero bidders or substituted with total bid counts.

Three linked ordinary least squares specifications will assess: (1) the association between opening and final prices conditional on product category and observable auction characteristics; (2) whether the low-opening-price association varies with seller reputation; and (3) whether it varies with distinct-bidder competition. The second and third specifications will include the corresponding interaction terms, with bidder competition mean-centered for interpretation. Models will use HC3 heteroskedasticity-robust standard errors. Sensitivity checks will include logarithmic price outcomes, the stricter under-pricing threshold, and category-specific estimates where sample sizes permit. The competition model will use only auctions with verifiable distinct-bidder information; observable differences between included and excluded listings will be summarized to assess possible selection bias. Because opening prices and bidder participation are not randomly assigned, these models estimate conditional associations, not causal effects. Variable definitions, exclusion criteria, and model specifications will be documented before the proposed extension is analyzed.

Research Topic

Strategic under-pricing is a familiar marketplace tactic, but its apparent simplicity hides an unresolved economic mechanism. A low opening price can attract attention, yet its success may depend on whether buyers trust the seller and whether enough independent bidders enter the auction. The topic is relevant to digital-market economics because

modern marketplaces combine pricing decisions, reputation systems, and strategic participation within the same transaction environment. The proposal is also methodologically useful: by defining under-pricing relative to product category and testing both threshold and continuous specifications, it distinguishes the pricing strategy itself from the possibility that expensive goods simply start and finish at higher prices. Testing reputation and competition in one newly constructed dataset can clarify when, if at all, a low opening price is associated with better auction outcomes.

References

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